
Attention Manipulation for Hallucination Mitigation

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Abstract

Recent advancements in large vision-language models have delivered outstanding results in tasks like image captioning and visual question-answering; yet these models still struggle with object hallucination, which undermines their reliability in safety-critical applications. In our work, we tackle this issue by exploring attention manipulation techniques: we first replicated the AtMan model to investigate its internal attention mechanisms and conducted ablation studies to evaluate the effect of modifying attention scores on model behavior. Building on this, we plan to improve the approach by incorporating bounding-box-based attention to improve the grounding of visual features and reduce hallucinations. We aim to show experimentally that this spatially aware strategy not only effectively reduces hallucination but also enhances the interpretability of the model, laying the groundwork for developing more robust and explainable LVLMs.

1 Introduction

Recent advances in large language models (LLMs) have driven significant progress in the field of large vision-language models (LVLMs), which aim to integrate sophisticated language comprehension with visual perception. These LVLMs have showcased impressive capabilities in tasks such as image captioning and visual question answering, enhancing the way machines perceive and interact with multimodal data. However, despite their successes, a major challenge persists: object hallucination. This phenomenon, where the model falsely generates or describes objects not present in an image, poses a significant reliability issue, particularly in critical real-world applications such as medical imaging, autonomous driving, and robotics [1, 2, 3].

Our work aims to address this challenge by investigating attention manipulation techniques, which are vital for capturing long-term dependencies and aligning textual and visual features. Specifically, we seek to answer the core research question: how can attention manipulation help mitigate object hallucination in LVLMs, thereby improving model reliability and interpretability?

To address this problem, we replicate and extend the AtMan model, which is a powerful language-model-based method that explains contents of the input image with simple prompts accurately and efficiently, as introduced in the core paper by Deiseroth et al. [4]. This novel approach has shown promising results in explaining how different visual components are interpreted by the model, offering valuable insights into model behavior. For our project, we replicated the AtMan model architecture, focusing on its attention manipulation mechanisms and conducting ablation studies to assess the impact of manipulating attention scores on model performance, which was a key area of interest for us during replication.

Through attention manipulation, we aim to identify and reduce instances where the model might hallucinate objects by modifying how attention is allocated to different parts of an input image. By

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controlling the attention flow, we can potentially guide the model to focus more on actually present visual features and reduce the likelihood of generating false descriptions.

While our current progress report focuses on single-object attention manipulation, our complete project will extend this approach to multi-object scenarios, which present additional challenges due to complex spatial relationships between objects. We plan to develop techniques that can handle multiple objects simultaneously, addressing potential cross-object hallucination issues.

Our approach offers a promising path toward more reliable and interpretable LVLMs that can accurately perceive and describe visual content without hallucinating non-existent objects, thus enhancing their utility in critical applications where accuracy is paramount.

2 Related Work

The application of explainable AI (XAI) is gaining increased attention in the AI community, particularly in advancing novel application domains. Longo et al. [5] emphasize the importance of XAI’s ability to apply existing methods in new contexts, proposing a manifesto that outlines several key propositions for guiding scientific research in XAI. Their work sets the foundational vision for expanding XAI beyond traditional tasks, thereby highlighting the growing need for explainable systems in emerging AI applications. Ukwaththa et al. [6] focus specifically on the use of XAI in 3D printing, detailing its role in design optimization, process monitoring, and sustainability analysis. Their work showcases the applicability of XAI to industrial processes, demonstrating its versatility in optimizing complex workflows. This work indicates the broader impact of explainable models beyond conventional AI research, particularly in improving transparency in automated manufacturing processes. Tornede et al. [7] explore the use of explainable methods in large language models (LLMs) and AutoML, emphasizing the importance of safeguards against the misuse of these powerful models. Their findings underline the challenge of ensuring safety in the deployment of advanced generative models and provide critical insights into the vulnerabilities of generative systems, which is highly relevant to our investigation of hallucination in LVLMs.

In the context of object hallucination in LVLMs, Wang et al. [8] propose an LLM-free evaluation pipeline designed to cater to different types of hallucination and tasks. However, their approach does not fully address the impact of object class distribution within an image during evaluation. This limitation is specifically addressed by the ROPE benchmark[9], which is why we integrated ROPE into our extension of the AtMan model. The ROPE benchmark provides a rigorous evaluation of multi-object scenarios, allowing us to better assess and mitigate hallucination behaviors in LVLMs. Our work builds upon these recent advances by situating our contributions in both explainability and mitigation of object hallucination.

3 Proposed Method

To enhance visual explainability, our work extends the AtMan architecture [4], which employs an autoregressive attention manipulation approach to elucidate visual features. AtMan acts as our foundational model for replication and further development. Rather than directly adopting the *MAGMA* pipeline [10] utilized by AtMan, we integrate the state-of-the-art LLaVA 1.5 model [11, 12] for improved performance. Additionally, we leverage the ROPE benchmark [9] to evaluate and enhance the model’s capability in managing hallucinations within multi-object scenarios.

3.1 Method Details

In our methods, attention is first introduced in [13] as follows:

$$\mathbf{O} = \text{softmax}(\mathbf{H}) \cdot \mathbf{V},$$

where $\cdot$ denotes matrix multiplication, $\mathbf{V}$ stands for the value matrix and $\mathbf{H}$ represents pre-softmax query-key attention scores. These pre-softmax attention scores are computed as:

$$\mathbf{H} = \mathbf{Q} \cdot \mathbf{K}^\top / \sqrt{d_h}.$$

where d_h is the dimensionality of each attention head. Since most contemporary AI models adopt an autoregressive structure, a lower-left triangular mask $\mathbf{M}$ is applied to enforce causality:

$$\mathbf{H}_M = \mathbf{H} \circ \mathbf{M},$$

with $\circ$ the Hadamard product. In practice, the masked attention score tensor has dimensions:

$$\mathbf{H}_M \in R^{h \times s \times s},$$

where h denotes the number of attention heads, and s represents the length of the input sequence. Intuitively, the attention scores quantify:

How does the model attend to different parts of the input sequence to predict certain outputs?

Consequently, we argue that these attention scores play a critical role in how the model interprets input information and formulates its predictions. Thus, visualizing and manipulating these attention scores can help uncover the underlying causes and inform potential mitigation strategies for hallucinations in VLMs.

3.2 Single Token Attention Manipulation

We follow the methods outlined in AtMan [4] and suppress individual token attention scores. For a token i in the input sequence, we apply a damping factor $\mathbf{f}$, modifying the attention scores across all layers and attention heads as:

$$\tilde{\mathbf{H}}_{u,*,*} = \mathbf{H}_{u,*,*} \circ (1 - \mathbf{f}^i),$$

where 1 denotes matrix containing ones and $\mathbf{f}^i$ contains the damping factors matrix. Intuitively, these factors decrease the model’s attention to token i , thus potentially influencing the model’s behavior.

To quantify how attention manipulation of a specific token affects the model’s overall predictions, we adopt the positional influence functions from AtMan [4]. We define the explanation vector for a target class label target as:

$$\mathcal{E}(\mathbf{w}, \text{target}) := (\mathcal{I}^{\text{target}}(\mathbf{w}_1, \mathbf{w}), \dots, \mathcal{I}^{\text{target}}(\mathbf{w}_n, \mathbf{w})),$$

with each positional influence function defined as:

$$\mathcal{I}^{\text{target}}(\mathbf{w}_i, \mathbf{w}) := L^{\text{target}}(\mathbf{w}, \theta_{-i}) - L^{\text{target}}(\mathbf{w}, \theta). \quad (1)$$

where θ_{-i} represents the model with the i th token suppressed and $\mathbf{w} = [w_1, \dots, w_N]$ is the input sequence with length N .

The loss function $L^{\text{target}}(\mathbf{w}, \theta)$ is defined as the negative log-softmax of the model’s predictions. Specifically, this is the negative log-likelihood loss, converting logits into probabilities through the softmax function, where minimizing this loss corresponds to maximizing prediction accuracy. Formally, this loss is expressed as:

$$L^{\text{target}}(\mathbf{w}, \theta) = - \max_{\theta} \log p_{\theta}(\mathbf{w})_{\text{target}^t} = - \sum_t \log p_{\theta}(w_t | w_{<t})_{\text{target}^t} \quad (2)$$

where target^t denotes the vocabulary index of the t th target token.

In simpler terms, we compute the average cross-entropy of the input sequence across all target tokens and evaluate how it changes when the i th token is suppressed compared to the original sequence. This explanation corresponds to a difference vector that captures the effects of all possible positional perturbations within the sequence, requiring n forward passes to compute.

3.3 Pipeline and Evaluation Setup

The pipeline of our attention manipulation approach is illustrated in Figure 1. In this architecture, the input image and text are first processed by their respective encoders - the visual encoder for images and the text encoder for textual input. These encoded representations are then fed into the Large Vision Model, which contains multiple attention layers. Our key contribution is the enhanced attention mechanism shown by the curved arrows, where we selectively increase attention weights within the bounding boxes of detected objects. This targeted attention manipulation guides the model to focus more precisely on actual objects present in the image, reducing the likelihood of

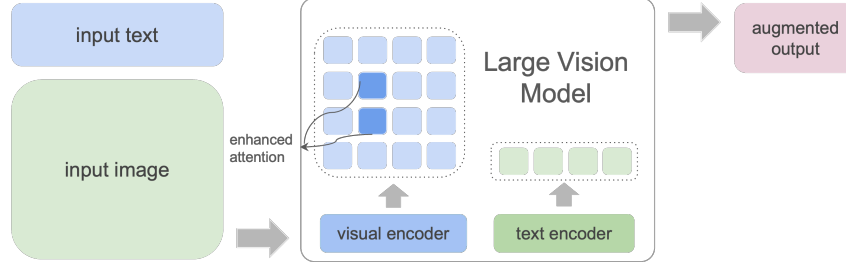


Figure 1: Pipeline of the proposed method. The model takes both text and image as inputs and produces outputs based on enhanced attention.

hallucinations. The model then produces an augmented output with improved object grounding and reduced false detections.

Following the setup in AtMan [4], we use the *OpenImages* dataset [14] as image inputs. OpenImages is a comprehensive dataset encompassing various image-level and pixel-level tasks. The dataset contains 41,600 images in its validation set, with bounding box annotations for object detection tasks. For our use case we adopt the vision QA task for evaluation.

We prompt our VLM using the approach similar to that in the AtMan Paper, where the problem is formulated as sentence completion. Due to the instruction following nature of our model, we prompt the *Llava-1.5-7b* model using the template:

User: What is in this picture?
Assistant: This is a picture of...

And then calculate the influence score as explained in Section 3.2. After that, we take the segmentation annotations M in the dataset as the ground truth and employ a varying cutoff to the influence scores to obtain a list of images patches $I_1, I_2, I_3, \dots, I_n$. Eventually, the model generates the augmented output. Therefore, we define the mean average precision/recall to be:

$$\text{mAP} = \text{mean} \left(\frac{(I_1 \cup I_2 \cup \dots \cup I_n) \cap M}{I_1 \cup I_2 \cup \dots \cup I_n} \right)$$

$$\text{mAR} = \text{mean} \left(\frac{(I_1 \cup I_2 \cup \dots \cup I_n) \cap M}{M} \right)$$

The mAP and mAR are calculated based on image tokens, since AtMAN method operates on the token granularity. Note that the mean value comes from varying the cutoff and obtaining different influential image patches.

4 Experiment Results

4.1 Overview

We evaluate our methods on both single-object and multi-object settings. In Sections 4.2 and 4.3, we conduct the single-object evaluation by replicating the AtMan paper [4] with our selected dataset and integrating the YOLO method. In Section 4.4, we experiment on multi-object scenarios by evaluating the ATMAN-in-Box method on the ROPE pipeline [9].

4.2 Replicating Attention Manipulation in AtMan

Figure 2 illustrates an example saliency map generated via our attention manipulation process, while Table 1 summarizes our initial results from adjusting single-token attention scores. In our experiments using the single-object subset of *OpenImages* validation set, we obtained nontrivial precision and recall metrics—specifically, an mAP of 68.58 (with an inter-quartile value of 68.72) and an mAR of 48.68 (IQ: 50.28).

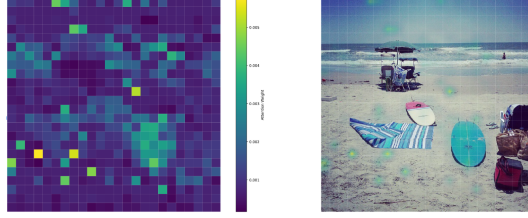


Figure 2: Visualization of AtMAN method as a saliency map. The lighter means the more important the token is for predicting the target token.

Metrics	Normal	IQ
mAP	68.58	68.72
mAR	48.68	50.28

Table 1: mAP and mAR of AtMan method on *OpenImages* dataset. *IQ* indicates inter-quartile values. The values are estimated on the first 100 data samples on the validation set.

Although these values are respectable, they do not completely match the results reported in the original AtMan paper [4]. We attribute these discrepancies primarily to two factors.

- One is the model differences in **model architecture** and **fine-tuning procedure**. We employed the *Llava-1.5-7b* model, which contrasts with the *MAGMA* model used in the original work. Firstly, *MAGMA* integrates adapter blocks within its transformer layers to capture additional visual cues, while LLaVA relies on a standard LLaMA backbone. Secondly, *MAGMA* is optimized solely for caption prediction, while LLaVA is instruction fine-tuned. These discrepancies between the models could largely shift the explanation generated by AtMan method.
- The other is the difference in data size. Our evaluation was conducted on a significantly smaller subset of *OpenImages* dataset (the first 100 data samples), and variations in sampling strategy may also influence the metric values.

4.3 YOLO Integration for Improved Bounding Box Proposals

Additionally, we observed that many of the ground truth bounding box in the *OpenImages* dataset were of low quality, as shown in the example in Figure 3. Therefore, we integrated YOLOv8 [15] for detecting objects and incorporating bounding boxes to guide attention manipulation. We randomly selected 100 images from the *OpenImages* and relabeled the bounding boxes. The resulting single-object evaluation from adjusting single-token attention scores is summarized in Table 2.

In the meantime, however, we observe that the YOLOv8 model has difficulty correctly identifying object classes. For instance, in Figure 4, the YOLOv8 pipeline (depicted in blue) mistakenly classifies

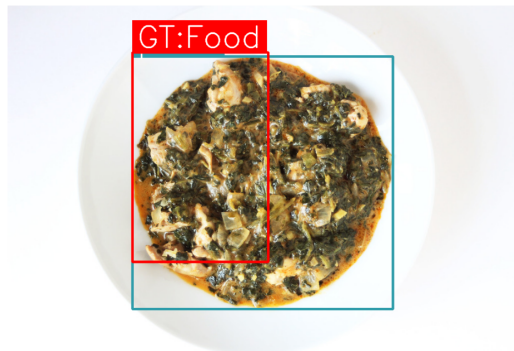


Figure 3: Visualization of *OpenImages* ground truth bounding box (red) and YOLO bounding box (teal).

an insect as a bird. To address this issue, we adopt the bounding boxes of the YOLOv8 model while preserving the class labels in the ground-truth dataset throughout our experiments.

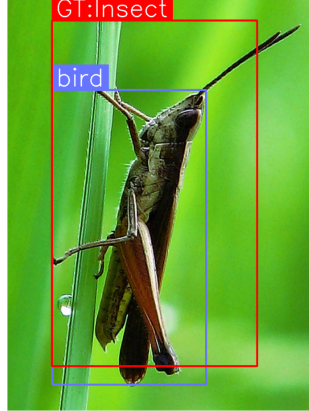


Figure 4: Classifications of *OpenImages* ground truth (red) and YOLO pipeline (blue).

Metrics	Normal	IQ
mAP	69.82	69.63
mAR	52.39	52.90

Table 2: mAP and mAR of AtMan method on *OpenImages* dataset with YOLOv8 bounding boxes. The values are estimated on the first 100 data samples on the validation set.

We see an improved mAP and mAR score across the board compared to the original dataset. Both qualitative comparison shown in Figure 3 and quantitative comparison between Table 1 and Table 2 suggest that our YOLOv8 pipeline provides bounding box regions that better captures object location and thus more faithfully measure whether the VLM is attending to correct locations on the image.

4.4 ATMAN-in-Box

We further expand our experiment to test spatially aware attention modulation, which we term **ATMAN-in-Box**. Specifically, we integrated bounding box information to adjust the attention weights, amplifying annotated object regions while dampening otherwise.

We test our method on the ROPE benchmark [9], which tests the VLM in the multi-object case. We prompt the *Llava-1.5-7b* model using the template:

User: Provide the class names in the format: 'obj1: <class1>, obj2: <class2>, obj3: <class3>, obj4: <class4>, obj5: <class5>', with no additional words or punctuations.

As summarized in Table 3, the accuracy increases as our ATMAN-in-Box method improves the performance of models in all cases, proving to be a promising attention modulation to reduce multi-object hallucination.

Model	Seen Wild	Seen Hom.	Seen Het.	Unseen Wild	Unseen Hom.	Unseen Het.
LLaVA-1.5	21.26%	52.40%	7.69%	13.96%	31.88%	3.98%
+ ATMAN-in-Box	23.80%	55.60%	8.50%	16.00%	35.60%	4.50%

Table 3: Evaluation results on the ROPE benchmark with the ATMAN-in-Box method.

In addition to the original images from the ROPE benchmark, we created a dataset with 100 images from *OpenImages* following the YOLOv8 pipeline discussed in the previous section, but instead of taking only one bounding box, we took the top-5 bounding boxes and labeled them $\text{obj}\{i\}$ with i

being 1 to 5 to meet the requirement of ROPE. All bounding boxes are of the same color to ensure fairness, as shown in Figure 5.

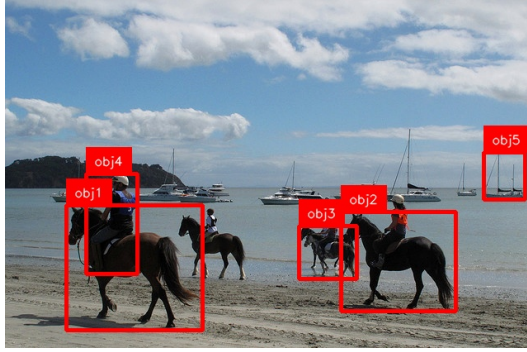


Figure 5: Example of the contributed YOLOv8 dataset.

We show that with the YOLO integration, the model demonstrates a significantly higher accuracy, as in Table 4. This demonstrates the high quality and high precision of our new dataset.

Model	Accuracy
LLaVA-1.5	21.26%
+ YOLOv8	41.43%

Table 4: Evaluation results on the ROPE benchmark with the YOLOv8 data.

4.5 Limitation and Future Work

Both the ATMAN-in-Box and YOLO integration demonstrate promising improvements in reducing multi-object hallucination. However, we observed a decrease in accuracy when these two methods were combined, resulting in an accuracy of only 26.00%. While this result is still better than the vanilla LLaVA result of 21.26%, it is lower than the reported accuracy of 41.43% when using the YOLO method without ATMAN-in-Box. This decrease may be attributed to the larger bounding box sizes generated by YOLO, as illustrated in Appendix B, which could make the model less sensitive to attention-amplified areas. To address this issue, future work should focus on developing an adaptive attention amplification method.

5 Conclusion

In this report, we introduced **ATMAN-in-Box** which mitigates object hallucination in large vision-language models through attention manipulation techniques. Our work began with replicating the AtMan model, where we carefully analyzed and modified single-token attention scores to understand their influence on model predictions. Upon observing low quality bounding boxes, we implemented a data labeling pipeline with YOLOv8 to create better data annotation on the *OpenImages* dataset. Our experiments demonstrated promising results on key accuracy metrics, underscoring the potential of our approach to enhance visual feature grounding and overall model interpretability.

Building on this foundation, we designed our ATMAN-in-Box extension, which leverages bounding box information to reinforce spatially-aware attention adjustments, amplifying focus on genuine object regions while reducing the influence of background noise. By evaluating on the ROPE benchmark, we showed that our method mitigates multi-object hallucination.

References

- [1] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36, 2024.
- [2] Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. Object hallucination in image captioning. *arXiv preprint arXiv:1809.02156*, 2018.

- [3] Wenliang Dai, Zihan Liu, Ziwei Ji, Dan Su, and Pascale Fung. Plausible may not be faithful: Probing object hallucination in vision-language pre-training. *arXiv preprint arXiv:2210.07688*, 2022.
- [4] Björn Deiseroth, Mayukh Deb, Samuel Weinbach, Manuel Brack, Patrick Schramowski, and Kristian Kersting. ATMAN: Understanding transformer predictions through memory efficient attention manipulation. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.
- [5] Luca Longo, Mario Brcic, Federico Cabitza, Jaesik Choi, Roberto Confalonieri, Javier Del Ser, Riccardo Guidotti, Yoichi Hayashi, Francisco Herrera, Andreas Holzinger, et al. Explainable artificial intelligence (xai) 2.0: A manifesto of open challenges and interdisciplinary research directions. *Information Fusion*, 106:102301, 2024.
- [6] Jeewanthi Ukwaththa, Sumudu Herath, and DPP Meddage. A review of machine learning (ml) and explainable artificial intelligence (xai) methods in additive manufacturing (3d printing). *Materials Today Communications*, page 110294, 2024.
- [7] Alexander Tornede, Difan Deng, Theresa Eimer, Joseph Giovanelli, Aditya Mohan, Tim Ruhkopf, Sarah Segel, Daphne Theodorakopoulos, Tanja Tornede, Henning Wachsmuth, et al. Automl in the age of large language models: Current challenges, future opportunities and risks. *arXiv preprint arXiv:2306.08107*, 2023.
- [8] Junyang Wang, Yuhang Wang, Guohai Xu, Jing Zhang, Yukai Gu, Haitao Jia, Ming Yan, Ji Zhang, and Jitao Sang. An llm-free multi-dimensional benchmark for mllms hallucination evaluation. *arXiv preprint arXiv:2311.07397*, 2023.
- [9] Xuweiyi Chen, Ziqiao Ma, Xuejun Zhang, Sihan Xu, Shengyi Qian, Jianing Yang, David Fouhey, and Joyce Chai. Multi-object hallucination in vision language models. In *3rd Workshop on Advances in Language and Vision Research (ALVR)*, 2024.
- [10] Constantin Eichenberg, Sidney Black, Samuel Weinbach, Letitia Parcalabescu, and Anette Frank. Magma—multimodal augmentation of generative models through adapter-based finetuning. *arXiv preprint arXiv:2112.05253*, 2021.
- [11] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning, 2023.
- [12] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *NeurIPS*, 2023.
- [13] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2023.
- [14] Ivan Krasin, Tom Duerig, Neil Alldrin, Andreas Veit, Sami Abu-El-Haija, Serge Belongie, David Cai, Zheyun Feng, Vittorio Ferrari, and Victor Gomes. Openimages: A public dataset for large-scale multi-label and multi-class image classification., 01 2016.
- [15] Rejin Varghese and Sambath M. Yolov8: A novel object detection algorithm with enhanced performance and robustness. In *2024 International Conference on Advances in Data Engineering and Intelligent Computing Systems (ADICS)*, pages 1–6, 2024.

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The project code is available at github.com/jhhuangchloe/545Project, and the contributed *OpenImages* dataset with *.json* files for classification is available at huggingface.co/datasets/eecs545/eecs545project. Please refer to the repositories for details.

B Additional Bounding Box Samples

YOLOv8 tends to generate larger and more accurate bounding boxes compared to the ground-truth boxes from *OpenImages*, as shown in Figure 6.



Figure 6: More visualizations of *OpenImages* ground-truth bounding boxes (red) and YOLO bounding boxes (others).

C Contribution of Team Members

Jiahe Huang and Xuejun Zhang conducted literature reviews on hallucination datasets. Ran Qiu and Yao-An Yang conducted extensive literature reviews to propose novel research directions, provided critical feedback on dataset suitability for hallucination detection. Tian Xia and Yao-An Yang explored foundational concepts of VLMs, implemented LLaVA using Hugging Face repositories, reviewed the MAGMA model literature. Jiahe Huang and Xuejun Zhang implemented attention manipulation techniques via AtMan, investigated explainability methods, and integrated visual explainability approaches into the MOH evaluation pipeline. Tian Xia, Jiahe Huang, Yao-An Yang, and Ran Qiu integrated YOLO into the data labelling. Xuejun Zhang conducted the ROPE experiments. All co-authors participated in preparing and writing this report. All co-authors contributed equally to this project.